

Solving Quadratics by Factoring

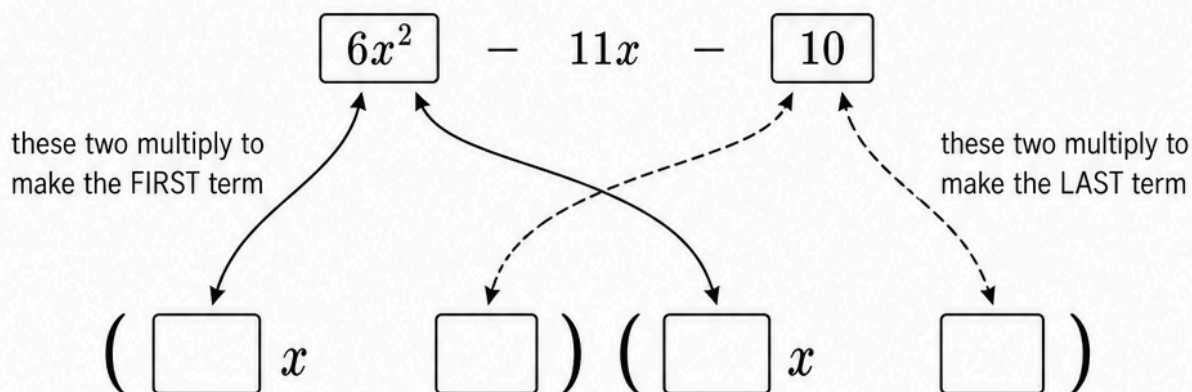
I. Trinomials

1. Start by looking for a GCF.

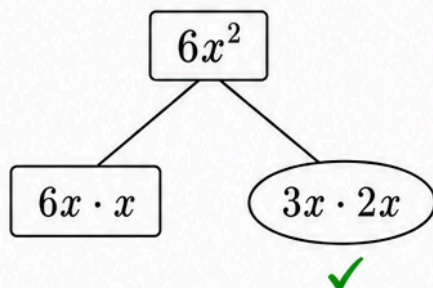
$$12x^2 - 22x - 20 = 0 \xrightarrow{\text{divide by 2}} 6x^2 - 11x - 10 = 0$$

Every term divides by 2, and the equation equals zero, so divide both sides. The 2 is gone for good.

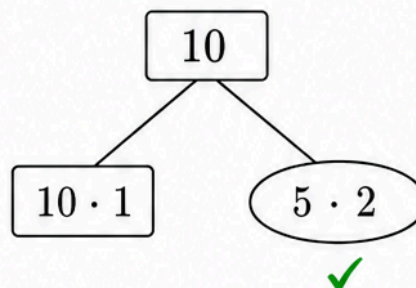
2. A trinomial factors into two sets of parentheses.



3. How can we make $6x^2$?



4. How can we make 10?



Always try the closest pair first.

5. Put the $3x$ and the $2x$ in the first spots.

It does not matter which order you put these two in.

$$(3x \quad \quad) (2x \quad \quad)$$

6. Now place the 5 and the 2.

Here the order does matter.

Option A $(3x \quad 5) (2x \quad 2)$

Option B $(3x \quad 2) (2x \quad 5)$

Factoring a Trinomial

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First try: put the 5 in the first set.

$$\left(\boxed{3}x + \boxed{5} \right) \left(\boxed{2}x + \boxed{2} \right)$$

First
 $3 \cdot 2 = 6$



$$\begin{array}{r} \text{inner} \quad 10x \\ \text{outer} \quad + \quad 6x \\ \hline 16x \end{array} \quad \times$$

Last
 $5 \cdot 2 = 10$



$16x$ is not $11x$. Also $2x$ and 2 share a factor of 2 , and the GCF is already out, so this arrangement cannot work.

➔ **Swap the 5 and the 2.**

Second try: swap them.

$$\left(\boxed{3}x + \boxed{2} \right) \left(\boxed{2}x + \boxed{5} \right)$$

First
 $3 \cdot 2 = 6$



Can 15 and 4 make 11 ? Yes, $15 - 4 = 11$.

$$\begin{array}{r} \text{outer} \quad -15x \\ + \text{inner} \quad 4x \\ \hline -11x \end{array} \quad \checkmark$$

Last
 $2 \cdot 5 = 10$



Now the signs are decided.

The middle term is negative, so the 15 must be negative. The 15 came from $3x \cdot 5$, so the 5 takes the minus.

$$\left(\boxed{3}x + \boxed{2} \right) \left(\boxed{2}x - \boxed{5} \right)$$

$$(+2)(-5) = -10$$

Set each set of parentheses equal to zero.

two things multiplying to zero means one of them is zero

$$\begin{aligned} 3x + 2 &= 0 \\ x &= \frac{-2}{3} \end{aligned}$$

$$\begin{aligned} 2x - 5 &= 0 \\ x &= \frac{5}{2} \end{aligned}$$